

Class 28: Basic Statistics and Graphs

Example Data Set 1: Masses of Common Candy Bars (in grams)

40, 43, 38, 21, 43, 51, 42

- I. Mean (average): The sum of the numbers divided by the number of numbers.
- II. Median: The middle number when the numbers are ordered from least to greatest.
- III. Mode: The number that appears the most frequently in the data set.
- IV. Range: The biggest number minus the smallest number.
- V. 1st quartile and 3rd quartile: The median of the first half of the data and the median of the second half of the data.
- VI. Interquartile Range: The 3rd quartile minus the 1st quartile.
- VII. Outlier: A number that is unusually big or small.

Example Data Set 2: Horsepower of Small Displacement Sport Motorcycles.

26.7, 42.5, 20.6, 31, 35.1, 43.3

Find the mean, median, mode, range, and interquartile range of the following data sets. Then, draw a dot plot and a box and whisker plot above the dot plot.

- ① Wingspans of Harpy Eagles (centimeters): 191, 191, 202, 188, 188

② Lifespans of Houseflies (Days): 26, 28, 25, 25, 24, 26, 26, 25.

③ Calories in Hamburgers: 444, 385, 398, 399.

④ Clock Speed of Computer Processors (Gigahertz):

3.7, 3.9, 3.6, 3.9, 4.0, 3.7, 3.5, 4.1, 3.9

⑤ Asking Price of a Used Gibson SG Standard Guitar (\$):

1200, 1200, 1200, 1150, 1000, 1050

⑥ Top Speed of Popular Sports Cars (mph):

218, 240, 217, 241, 251, 205, 221

⑦ Billy's Video Game Scores:

7000, 7000, 7000, 5000, 7000, 7000, 6000, 4000, 8000, 4500

Bar Graphs

⑧ Favorite Food of Children in a 6th Grade Classroom:

Food	Pizza	Tacos	Hamburgers	Fish
# of children	10	7	4	1

⑨ Favorite Color of Children in a 6th Grade Classroom:

color	Blue	Green	Red	Yellow	Pink	Orange	Purple
# of children	7	1	4	2	5	0	3

⑩ Movie Ratings:

Rating	Poor	Good	Excellent
# of people	16	52	80

⑪ Hair Color of Students in a 6th Grade Class:

color	Brown	Black	Blond	Red
# of students	15	4	2	1

Line Graphs

⑫

Rainfall (inches)	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Month	0	0	0.1	0	0.5	1	4	8.5	11	2	1.3	0.8

⑬ Electric Car Sales (\$):

Year	2012	2013	2014	2015
\$ Sales (millions)	23	50	61	90

⑭ Total Textbook Weight by Grade (lbs):

Grade	1	2	3	4	5	6	7	8	9	10	11	12
Total Textbook Weight (lbs)	5	4	5	5	6	8	11	10	15	17	19	20

⑮ 600 Yard Sprint Times of a Runner:

Training Day	1	2	3	4	5	6	7	8	9
Times (seconds)	305	301	298	295	273	270	269	275	270